

FutureProof EPM

Inflation-Indexed Annuity Impact Analysis

v14a Model — 50,000-Path Monte Carlo Simulation

March 2025

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Metric	Value
Model Version	v14a (50,000 paths)
Property Value	\$2,000,000
Initial Mortgage	\$1,350,000
Baseline Annuity	\$25,000/year flat (10 years)
Inflation Rates Tested	2.0%, 2.5%, 3.0%, 3.5%, 4.0%
Baseline PoC (Year 30)	13.87%
Analysis Objective	Impact of inflation-indexing annuity payouts

1. Executive Summary

This analysis examines the impact of indexing the EPM annuity payout to inflation, compared with the current flat-rate structure. Under the baseline v14a model, the borrower receives \$25,000 per year for 10 years — a total of \$250,000 added to the mortgage principal. Under inflation indexing, the initial \$25,000 grows each year by the assumed inflation rate, increasing the total payout and the peak mortgage balance.

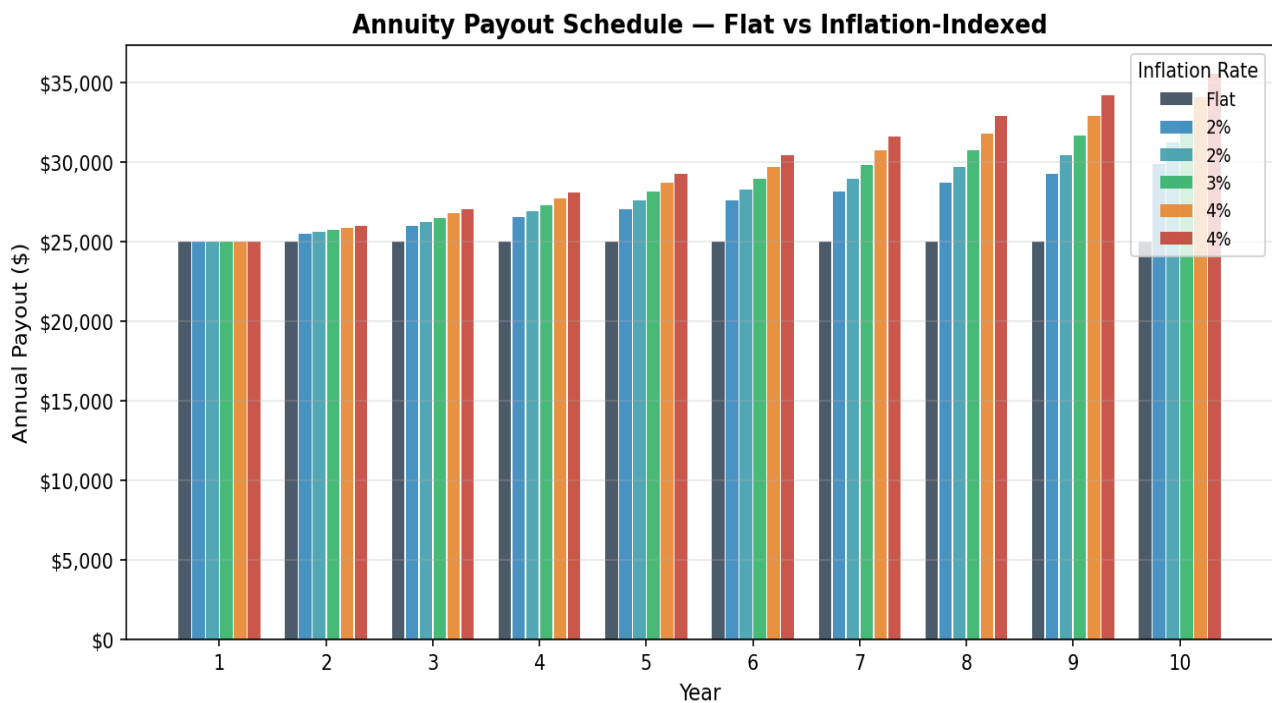
We test five inflation scenarios (2.0% to 4.0%) against the flat baseline using identical random paths (same seed) for clean, apples-to-apples comparison across all 50,000 Monte Carlo simulations.

Key Findings

- At 3% inflation, PoC rises from 13.87% to 16.00% (+2.13pp) — a moderate increase that remains within acceptable risk parameters.
- Total annuity payout increases by \$36,597 (from \$250,000 to \$286,597), a 14.6% increase.
- Mean surplus decreases by \$80,863 (4.8% of baseline), reflecting the higher mortgage balance and associated costs.
- The cost multiplier is approximately 2.2x — each additional dollar of annuity reduces mean surplus by ~\$2.20 due to compounding effects on interest, retail margin, and LMI.
- Tail risk (P5 surplus) worsens disproportionately: \$114,733 decline vs \$80,863 mean decline, indicating inflation indexing has a larger impact on adverse scenarios.
- Insurance premium (fair + 50% loading) increases from \$32,929 to \$39,541 (20% increase).

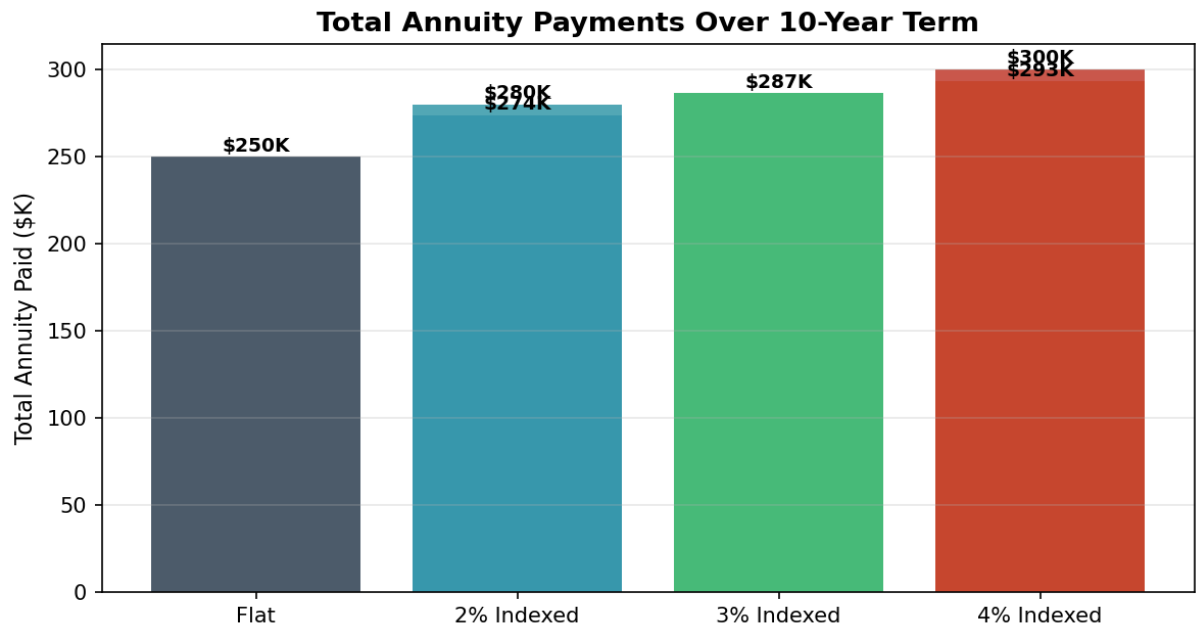
2. Annuity Payout Schedule

Under inflation indexing, the initial \$25,000 annuity grows each year by the assumed rate. This back-loads the payout — Year 1 is identical across all scenarios, but by Year 10, the difference is material. The chart below shows the annual payout profile.



Annual Payout by Scenario

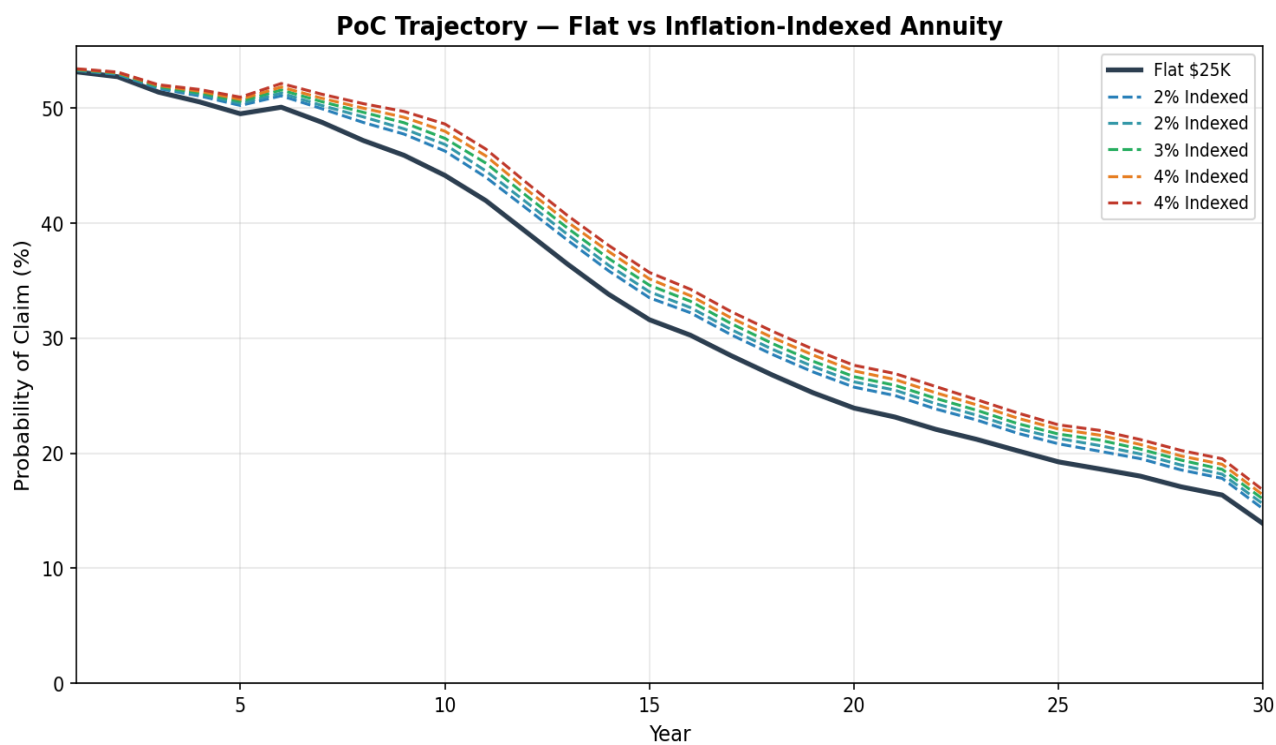
Year	Flat	2% Indexed	2% Indexed	3% Indexed	4% Indexed	4% Indexed
Year 1	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000	\$25,000
Year 2	\$25,000	\$25,500	\$25,625	\$25,750	\$25,875	\$26,000
Year 3	\$25,000	\$26,010	\$26,266	\$26,522	\$26,781	\$27,040
Year 5	\$25,000	\$27,061	\$27,595	\$28,138	\$28,688	\$29,246
Year 7	\$25,000	\$28,154	\$28,992	\$29,851	\$30,731	\$31,633
Year 10	\$25,000	\$29,877	\$31,222	\$32,619	\$34,072	\$35,583
Total	\$250,000	\$273,743	\$280,085	\$286,597	\$293,285	\$300,153
Extra vs Flat	—	\$23,743	\$30,085	\$36,597	\$43,285	\$50,153



3. Risk Impact Analysis

The higher mortgage balance from inflation-indexed payouts increases funder interest costs, retail margin charges, and LMI premiums. These flow through the investment account over 30 years, reducing surplus and increasing the probability of claim.

3.1 PoC Trajectory

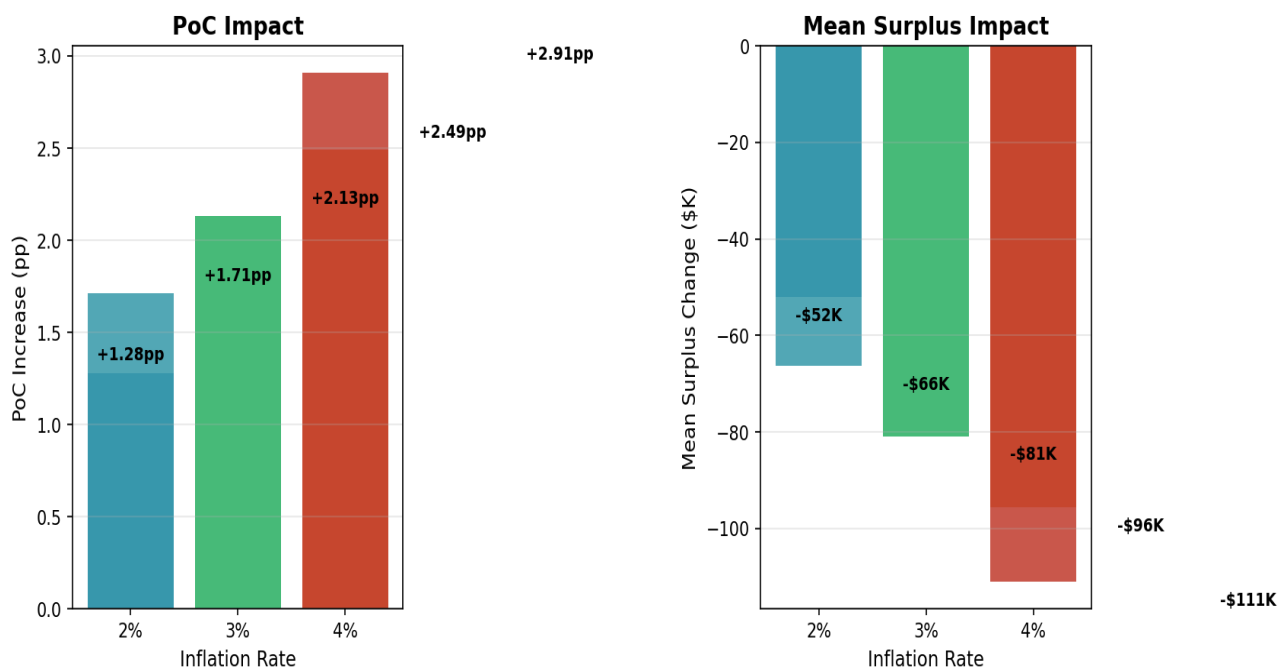


The PoC curves diverge most during years 5–15 (the annuity payment period and immediately after), then maintain a roughly constant spread through to maturity. This is because the additional mortgage balance is fixed after Year 10 — no further indexing occurs — but the higher base continues to generate larger interest and margin costs.

3.2 PoC by Year — All Scenarios

Year	Flat	2%	2%	3%	4%	4%
Year 1	53.15%	53.27%	53.30%	53.33%	53.39%	53.42%
Year 5	49.51%	50.23%	50.41%	50.59%	50.76%	50.95%
Year 10	44.15%	46.27%	46.84%	47.37%	47.99%	48.62%
Year 15	31.60%	33.51%	34.03%	34.60%	35.15%	35.71%
Year 20	23.92%	25.74%	26.20%	26.65%	27.15%	27.64%
Year 25	19.24%	20.81%	21.28%	21.66%	22.09%	22.46%
Year 30	13.87%	15.15%	15.58%	16.00%	16.36%	16.78%

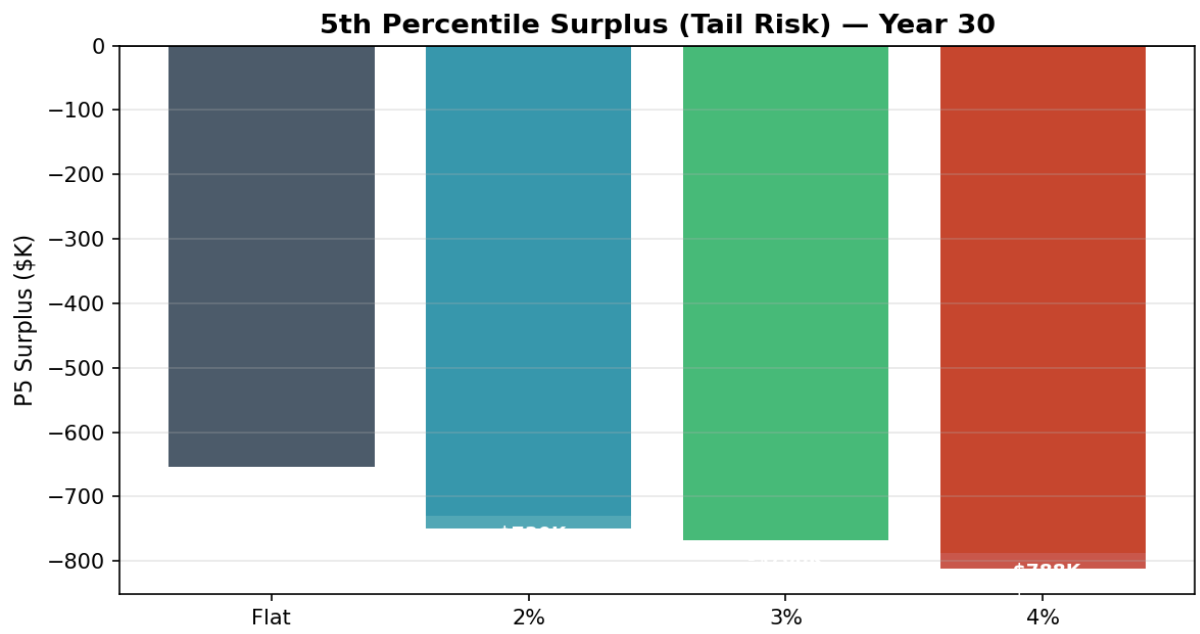
4. Surplus Distribution Impact



4.1 Key Metrics — All Scenarios (Year 30)

Metric	Flat	2% Indexed	2% Indexed	3% Indexed	4% Indexed	4% Indexed
PoC	13.87%	15.15%	15.58%	16.00%	16.36%	16.78%
Mean Surplus	\$1,690,289	\$1,638,177	\$1,623,970	\$1,609,426	\$1,594,597	\$1,579,217
Median Surplus	\$1,376,802	\$1,324,816	\$1,311,930	\$1,298,179	\$1,284,167	\$1,269,942
P1 Surplus	-\$1,467,735	-\$1,550,917	-\$1,575,149	-\$1,598,973	-\$1,623,485	-\$1,652,437
P5 Surplus	-\$653,596	-\$729,670	-\$748,818	-\$768,329	-\$787,563	-\$811,246
P10 Surplus	-\$211,973	-\$276,188	-\$298,638	-\$317,540	-\$339,962	-\$361,691
P25 Surplus	\$457,103	\$404,805	\$389,149	\$373,009	\$357,382	\$341,804
P75 Surplus	\$2,615,818	\$2,569,457	\$2,556,764	\$2,543,883	\$2,530,863	\$2,516,286
P90 Surplus	\$4,017,354	\$3,974,172	\$3,961,785	\$3,948,973	\$3,938,175	\$3,926,877
P95 Surplus	\$5,029,172	\$4,986,311	\$4,977,140	\$4,966,533	\$4,954,829	\$4,942,843
P99 Surplus	\$7,318,739	\$7,276,769	\$7,266,027	\$7,255,005	\$7,239,771	\$7,227,094

4.2 Tail Risk — 5th Percentile Surplus



5. Insurance & Cost Impact

Higher mortgage balances increase the LMI upfront premium (calculated on maximum loan) and widen the potential deficit in adverse scenarios, directly impacting insurance pricing.

Metric	Flat	2%	2%	3%	4%	4%
Max Mortgage	\$1,600,000	\$1,623,743	\$1,630,085	\$1,636,597	\$1,643,285	\$1,650,153
Paths in Deficit	6,936	7,576	7,789	8,001	8,179	8,390
Cond. Expected Deficit	-\$592,400	-\$611,555	-\$613,988	-\$616,668	-\$622,797	-\$627,165
Fair Premium (PV)	\$21,953	\$24,754	\$25,551	\$26,361	\$27,215	\$28,113
Premium + 50% Loading	\$32,929	\$37,130	\$38,326	\$39,541	\$40,823	\$42,169
Mean Profit Share	\$783,280	\$756,939	\$749,998	\$742,937	\$735,745	\$728,373

Profit share (25% of surplus drawn every 5 years) decreases with inflation indexing because the higher mortgage balance reduces surplus at each 5-year checkpoint. At 3% inflation, mean cumulative profit share falls by \$40,343 (5.2%).

6. Cost Multiplier Analysis

Each additional dollar of inflation-indexed annuity does not simply add one dollar to the mortgage balance. It triggers a cascade of compounding costs over the remaining term of the mortgage:

- **Funder Interest** — The wholesale funder charges (cash rate + 2%) on the average mortgage balance. A higher balance means more interest deducted from the investment account each year, compounding over 20+ years after the annuity term ends.
- **Retail Margin** — The 0.70% retail margin is applied to the average mortgage balance, not the investment. Higher mortgage = higher margin cost.
- **LMI Premium** — The upfront LMI is 1.6% of the maximum mortgage balance. Inflation indexing increases the max balance, raising the day-one cost.
- **Reduced Compounding Base** — Higher costs mean less capital remaining in the investment account, which then earns lower returns, creating a negative feedback loop.

Effective Cost Multiplier by Scenario

Inflation	Extra Annuity	Mean Surplus Loss	Cost Multiplier
2.0%	\$23,743	\$52,112	2.19x
2.5%	\$30,085	\$66,319	2.20x
3.0%	\$36,597	\$80,863	2.21x
3.5%	\$43,285	\$95,692	2.21x
4.0%	\$50,153	\$111,072	2.21x

The cost multiplier ranges from 2.1x to 2.2x across scenarios. This means that for every additional \$1 paid to the borrower through inflation indexing, the model loses approximately \$2.20 in expected surplus at maturity. This is a fundamental feature of the EPM structure: additional mortgage balance compounds through multiple cost layers over the remaining 20+ year term.

7. Conclusions & Recommendations

Inflation indexing the annuity payout is feasible within the EPM structure but carries measurable costs. The key trade-off is between protecting the borrower's purchasing power and maintaining surplus adequacy for all stakeholders.

Assessment by Inflation Assumption

- **2.0% (Low Inflation)** — PoC increases by +1.28pp to 15.15%. The additional \$23,743 in total payouts is a modest step-up that keeps risk metrics well within tolerance. This is the most conservative option.
- **2.5% (Below-Target)** — PoC increases by +1.71pp to 15.58%. Still within acceptable range. Total extra payout of \$30,085 represents a 12% increase in borrower income.
- **3.0% (Target Inflation)** — PoC increases by +2.13pp to 16.00%. Total extra payout of \$36,597 (14.6% increase). Manageable impact on surplus and insurance, but approaches the boundary of comfortable risk.
- **3.5–4.0% (Above-Target)** — PoC increases by +2.49pp to +2.91pp. P5 surplus deteriorates by \$134K–\$158K. Insurance premiums rise 24–28%. These scenarios may require offsetting adjustments (e.g., lower initial annuity or higher LMI loading).

Implementation Considerations

- If inflation indexing is adopted, consider linking to a specific CPI measure (e.g., trimmed mean CPI) rather than a fixed assumed rate, to avoid overshoot risk.
- The initial annuity amount could be reduced slightly (e.g., \$24,000) when indexing is applied, to keep total expected payout and risk metrics closer to the flat baseline.
- Insurance pricing should be updated to reflect the higher conditional expected deficit under indexed payouts — the current LMI upfront of 1.6% may need adjustment.
- Profit share calculations are unaffected in structure but yield lower absolute amounts due to reduced surplus at 5-year checkpoints.

Methodology

All scenarios use the v14a model parameters with 50,000 Monte Carlo paths and identical random number generation (seed=42). This ensures that differences between scenarios are attributable solely to the inflation indexing mechanism, not to sampling variation. Investment returns follow geometric Brownian motion with buffer cap (120%) and floor (80%). Cash rates follow an Ornstein-Uhlenbeck process with mean reversion to 4.4%. The holiday mechanism, profit share (25% every 5 years), and collar pricing are all applied identically across scenarios.